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Instrumentation of a wind turbine for telemetry in the context of the internet of things

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Abstract

This research project designs a telemetry system for remote monitoring of a low-power wind turbine with internet connectivity, enabling the quantification of several variables. It also designs and builds housings to protect the instrumentation. The MQTT protocol, a lightweight communication protocol, will be used for remote communication. Modal analysis will be used to obtain the three fundamental parameters (natural frequency, damping ratios, and deformation patterns) that will help us understand the wind turbine's vibration behavior. Sensors (accelerometers) will be installed on the wind turbine to collect data and analyze it in Matlab-Simulink.

Introduction

Currently, wind power systems are the most efficient green systems for electrical energy production. However, they have some disadvantages, such as unpredictable power generation, not being recommended in urban environments, and a decrease in efficiency over time, among others. It is also known that vibration, when present in engineering structures, generates fatigue and effects that can affect system behavior. Therefore, data collection (vibration amplitude, modal parameters, and structural integrity) is required, as well as analysis to understand how these parameters affect the behavior and overall performance of a structure. Due to this, it is proposed to remotely monitor the parameters of a low-power wind turbine in real time, specifically analyzing vibration, current, voltage, and revolutions per minute.

Methodology

The development of this research topic involves meeting several specific objectives. Some of these are:

1. Integrate existing modules to develop the necessary instrumentation to measure voltage, current, revolutions per minute, wind potential, and wind speed.
2. Study modal parameter identification techniques for analyzing the structural health of wind turbines, using and integrating mathematical and computational tools.
3. Select appropriate enclosures for the protection and galvanic and environmental isolation of the telemetry and sensor system.
4. Integrate a set of available software and hardware modules to implement the telemetry system using remote data transfer protocols (MQTT).

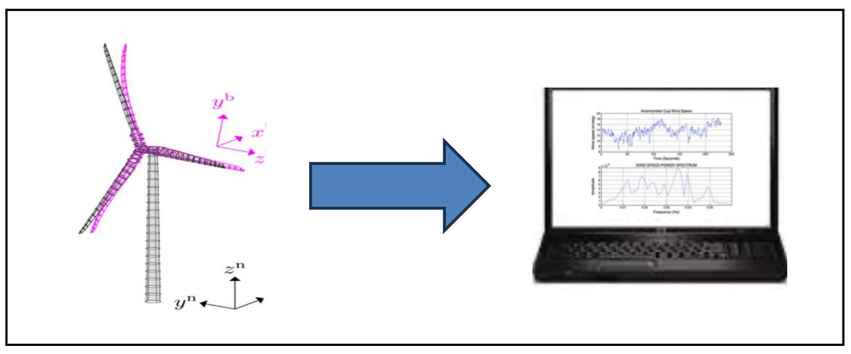

Fig.1. Sending data from the low-power wind turbine sensors to the software.

Results and discussion

It is known that the general equation that describes the motion of a one-degree-of-freedom system is the following:

$$m\ddot{x} + c\dot{x} + kx = 0 \qquad \text{Ecuation 1.}$$

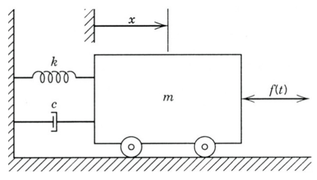

Fig.2. Example of a one-degree-of-freedom vibratory system.

With which you can obtain the behavior of the system, as well as the natural frequencies and the damping ratio.

$$w_n = \sqrt{\frac{k}{m}} \qquad \text{Ecuation 2.}$$

$$\zeta = \sqrt{1 - \left(\frac{w_d}{w_n}\right)^2} \qquad \text{Ecuation 3.}$$

For the wind turbine, the generalized system of a GDL is applied, which gives the following expressions:

$$(0.2267\rho AL + M)\ddot{z} + \left(\frac{\pi^4 EI}{32L^3} - \frac{\pi^2 Mg}{8L}\right) z = 0$$

with a natural frequency: $$w_n = \sqrt{\frac{\left(\frac{\pi^4 EI}{32L^3} - \frac{\pi^2 Mg}{8L}\right)}{(0.2267\rho AL + M)}}$$

On the other hand, instrumentation for data acquisition from the wind turbine was started, using a Raspberry PI 4, an Arduino UNO acquisition card, and an ESP8266 connected to an Adxl335 accelerometer. All transmission is carried out using the MQTT communication protocol.

Fig.3. Raspberry PI 4 card. Instrumentation (Raspberry Pi, Arduino, and accelerometer). Program for accelerometer data acquisition.

Conclusions

The project will provide a remote monitoring system for the operating conditions and structural parameters of a low-power wind turbine, allowing for a realistic analysis of its performance over time.

References

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Author

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